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LINEAR ACTUATOR SYSTEM FOR MOTION SIMULATOR

Abstract

A linear actuator system may have an actuator assembly for moving an output in translation in a first direction. A transmission has a frame, a joining link(s) pivotally connected to the frame at a first location and operatively connected to the actuator assembly at a second location for receiving movement from the output. The joining link(s) contacting an interface at a third location to cause relative movement between the frame and the interface in a second direction differing from the first direction. A motion platform system is also provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is a continuation of U.S. patent application Ser. No. 18/001,752, which is an application filed under 35 USC 371 of PCT Application No. PCT/CA2021/050814, filed on Jun. 25, 2021, which claims the benefit of U.S. Patent Application No. 63/039,078 filed on Jun. 15, 2020, and of U.S. Patent Application No. 63/165,319 filed on Mar. 24, 2021, the contents of both of which are incorporated herein by reference.

FIELD OF THE APPLICATION
[0002] The present application relates to linear actuators as used with motion simulators or in motion simulation, for instance to displace an occupant or occupants of a platform in synchrony with a sequence of video images or with an audio track, whether at home or in a theater, to watch movies, television, to play video games.
BACKGROUND OF THE ART
[0003] In the entertainment industry and in the gaming industry, there is an increasing demand for enhancing the viewing experience of a viewer or gamer. Accordingly, there has been numerous innovations to improve the image and the sound of viewings. Motion simulation has also been developed to produce movements of a motion platform (e.g., a seat, a chair) in synchrony with sequences of images of a viewing. For instance, U.S. Pat. Nos. 6,585,515 and 7,934,773 are two examples of systems that have been created to impart motion to a seat, to enhance a viewing experience.
[0004] Electro-mechanical linear actuators are commonly used in such motion platforms. These linear actuators must often be capable of producing low and medium amplitude outputs, at low or medium frequency, for a high number of strokes. Moreover, these linear actuators must support a portion of the weight of a platform and its occupant(s).
[0005] Linear actuators are typically elongated components that are positioned in a vertical orientation. This therefore imposes a constraint of height to motion platforms that use such vertically oriented actuators. It would be desirable to change an orientation of the linear actuators while not impacting substantially their performance.
SUMMARY OF THE APPLICATION
[0006] It is therefore an aim of the present disclosure to provide a linear actuator that addresses issues associated with the prior art.
[0007] It is a further aim of the present disclosure to provide a motion platform system that addresses issues associated with the prior art.
[0008] Therefore, in accordance with a first aspect of the present application, there is provided a linear actuator system comprising: an actuator assembly for moving an output in translation in a first direction; and a transmission having a frame, at least one joining link pivotally connected to the frame at a first location and operatively connected to the actuator assembly at a second location for receiving movement from the output, the at least one joining link contacting an interface at a third location to cause relative movement between the frame and the interface in a second direction differing from the first direction.
[0009] Further in accordance with the first aspect, for example, the first direction and the second direction are generally transverse to one another.
[0010] Still further in accordance with the first aspect, for example, the at least one joining link has the first location, the second location and the third location in a L pattern.
[0011] Still further in accordance with the first aspect, for example, the at least one joining link has a triangular shape.
[0012] Still further in accordance with the first aspect, for example, a pair of the at least one joining

link share a first rotational axis at the first location and share a second rotational axis at the second location.
[0013] Still further in accordance with the first aspect, for example, the pair share a third rotational axis at the third location.
[0014] Still further in accordance with the first aspect, for example, the at least one joining link is pivotally connected to the output of the actuator assembly at the second location.
[0015] Still further in accordance with the first aspect, for example, the at least one joining link is pivotally connected to at least one link at the third location, the at least one link being pivotally connected to the interface.
[0016] Still further in accordance with the first aspect, for example, the interface is pivotally connected to the frame.
[0017] Still further in accordance with the first aspect, for example, the interface has a pair of arms projecting from a central member, the pair of arms being pivotally connected to the frame, the central member pivotally connected to the at least one link.
[0018] Still further in accordance with the first aspect, for example, the at least one joining link is pivotally connected to at least a first link at the second location, the first link being pivotally connected to the output of the actuator assembly.
[0019] Still further in accordance with the first aspect, for example, the at least one joining link is pivotally connected to at least one second link at the third location, the second link being pivotally connected to the interface.
[0020] Still further in accordance with the first aspect, for example, the interface is pivotally connected to the frame, and the actuator assembly is secured to the frame.
[0021] Still further in accordance with the first aspect, for example, the interface has a pair of arms projecting from a central member, the pair of arms being pivotally connected to the frame, the central member pivotally connected to the second link.
[0022] Still further in accordance with the first aspect, for example, the frame defines a receptacle to receive at least a portion of the actuator assembly.
[0023] Still further in accordance with the first aspect, for example, actuator assembly is a linear actuator.
[0024] Still further in accordance with the first aspect, for example, the linear actuator is a bi-directional electromechanical linear actuator.
[0025] In accordance with a second aspect of the present disclosure, there is provided a motion platform system comprising: a support structure; a motion structure operatively mounted to the support structure by at least one joint so as to be displaceable relative to the support structure in at least one degree of freedom; and at least one of the linear actuator system as described above, the linear actuator system being between the support structure and the motion structure, the linear actuator system actuatable to impart movement to the motion structure in the at least one degree of freedom.
[0026] Further in accordance with the second aspect, for example, the motion structure includes a first panel configured to define a motion platform.
[0027] Still further in accordance with the second aspect, for example, the first panel has receptacles configured for receiving casters of a chair.
[0028] Still further in accordance with the second aspect, for example, the receptacles are elongated troughs.
[0029] Still further in accordance with the second aspect, for example, there are five of the elongated troughs, the elongated troughs being circumferentially distributed 72 degrees apart.
[0030] Still further in accordance with the second aspect, for example, there may be provided a strap for each receptacle.
[0031] Still further in accordance with the second aspect, for example, the first panel has a pentagonal shape.

[0032] Still further in accordance with the second aspect, for example, a second panel may be located under the first panel, the linear actuator system being fixed to the second panel.
[0033] Still further in accordance with the second aspect, for example, the at least one joint is connected to the second panel.
[0034] Still further in accordance with the second aspect, for example, the second panel has a pentagonal shape.
[0035] Still further in accordance with the second aspect, for example, the support structure is a third panel being located under the second panel.
[0036] Still further in accordance with the second aspect, for example, the third panel has a pentagonal shape.
[0037] Still further in accordance with the second aspect, for example, the support structure is the ground.
[0038] Still further in accordance with the second aspect, for example, a spherical joint may be between the linear actuator system and the support structure.
[0039] Still further in accordance with the second aspect, for example, the at least one joint has a pivot.
[0040] Still further in accordance with the second aspect, for example, the at least one joint includes two pivot members spaced apart and sharing a common rotational axis.
[0041] Still further in accordance with the second aspect, for example, a height between the support structure and a support plane of the motion structure is at most 12 inches high.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0042] FIG. **1** is a perspective view of a linear actuator system for motion simulators in accordance with the present disclosure;
[0043] FIG. **2** is a top view of the linear actuator system of FIG. **1**;
[0044] FIG. **3** is an elevation view of the linear actuator system of FIG. **1**;
[0045] FIG. **4** is an isometric view of a transmission group of the linear actuator system of FIG. **1**;
[0046] FIG. **5** is a top view of the transmission group of the linear actuator system of FIG. **1**;
[0047] FIGS. **6**A to **6**C are a sequence showing a conversion of movement from horizontal to vertical as permitted by the linear actuator system of FIG. **1**;
[0048] FIG. **7** is a perspective view of an embodiment of a motion platform system for motion simulators in accordance with an embodiment of the present disclosure, with outer portions of a motion platform thereof being see-through to show underlying portions of the motion platform system including the actuator of FIG. **1**;
[0049] FIG. **8** is an exploded view of the motion platform system of FIG. **7**;
[0050] FIG. **8**A is an exploded view of a first joint and of a bracket of the motion platform system of FIG. **7**;
[0051] FIG. **8**B is an exploded view of another joint of the motion platform system of FIG. **7**, with portions being see-through and/or exploded;
[0052] FIG. **9** is an elevation view of portions of the motion platform system of FIG. **7**;
[0053] FIG. **10** is a top view of the motion platform system of FIG. **7**;
[0054] FIG. **11** is a perspective view of another motion platform system in accordance with another embodiment of the present disclosure; and
[0055] FIGS. **12**A to **12**E are schematic side views of contemplated configurations of the linear actuator system of the present disclosure.
DESCRIPTION OF THE PREFERRED EMBODIMENTS
[0056] Referring to the drawings and more particularly to FIGS. **1** to **3**, there is illustrated at **10** a

linear actuator system of the type used for motion simulators. The linear actuator system **10** is well suited to be used between the ground or a baseplate and a motion platform (i.e., support surface, chair, seat, flight simulator/compartment, etc) to displace the motion platform in synchrony with a sequence of images and/or sound, for instance part of a motion picture, a televised event, a video, a video game, a simulation, haptic event, a virtual reality session, etc. The linear actuator system **10** of the illustrated embodiments is an electro-mechanical linear actuator that is driven by a motion controller, or any other appropriate and adapted source of motion signals (e.g., media player, D-cinema projector, internet, etc), e.g., code representing specific motions to be performed. The motion signal is sent to the linear actuator system **10** in a suitable format to drive a motor thereof. In an embodiment, one or more of the actuator system **10** are used concurrently to support and displace a seat relative to the ground (ground including a structure on the ground). The linear actuator system **10** therefore produces a translational output, along an axial direction thereof, illustrated as X, but the output is converted into a movement along Y. In an embodiment, direction X is generally horizontal in use, while direction Y is generally vertical. However, this is an option.

[0057] The linear actuator system **10** may be an assembly of four groups (i.e., portions, assemblies, sub-assemblies, etc), namely a motor group **20**, a structural group **30**, a driven group **40**, and a transmission group **50**. The expression “group” is used merely to simplify the following description. The motor group **20**, the structural group **30** and the driven group **40** are only schematically illustrated and briefly detailed, as the details of the present disclosure mostly pertain to the transmission group **50**. For reference, PCT application no. PCT/US2013/072605 describes one example of a motor group **20**, of a structural group **30**, and of a driven group **40** and is hence incorporated by reference. Components shown as being part of a group may be part of another group, may be shared by groups, etc.

[0058] The motor group **20** may receive motion signals in electric format, and may produce rotational motions corresponding to the motion signals received, as a possibility among others. In such an embodiment, the motor group **20** is therefore connected to a source of motion signals or like electronic equipment. The motor group **20** is operatively connected to the driven group **40** to transmit its rotational motions thereto. The motor group **20** may have an electric motor **21**. The electric motor **21** may be a bi-directional motor of the type receiving an electrical motion signal, to convert the signal in a rotational output proportional to the motion signal, in either circular directions, in direct drive. By way of example, the electric motor **21** is a brushless DC motor. This type of electric motor is provided as an example, and any other appropriate type of motor may be used. In alternative embodiments, instead of an electric motor, a pneumatic motor, an hydraulic motor, or cylinders are used to produce a reciprocating translational movement.

[0059] The structural group **30** may house at least part of the driven group **40**, and operatively connects the motor group **20** to the driven group **40**. Moreover, the structural group **30** may be the interface between the linear actuator system **10** and the motion platform, the ground, or a supporting structure. The structural group **30** may include a casing **31**, also known as a cover, housing, or the like. In the illustrated embodiment, the casing **31** is a monolithic piece. The casing **31** is a main structural component of the linear actuator system **10**, as it interfaces the motor group **20** to the driven group **40**, and may also interface the linear actuator system **10** to the transmission group **50**.

[0060] The driven group **40** converts the rotational motions from the motor group **20** into linear motions along direction X. The driven group **40** is displaceable relative to the structural group **30**, and is shown emerging out of the casing **31** in FIG. **1**.

[0061] Still referring to FIGS. **1** to **3**, a transmission group **50** is shown relative to a remainder of the linear actuator system **10**, i.e. relative to the motor group **20**, the structural group **30** and the driven group **40**, all of which are referred to as actuator assembly herein. In FIGS. **4** and **5**, the motor group **20**, the structural group **30** and the driven group **40** are removed for clarity and to better illustrate the transmission group **50** alone. The transmission group **50**, also known as a transmission assembly, transmission, etc, is tasked with changing a direction of the output from the actuator assembly, for example from direction X to direction y, or any other desired direction depending on the application. As a whole, the linear actuator system **10** pushes and pulls, but may also rely on gravity to assist in lowering the motion platform MP (FIGS. **6**A to **6**C).

[0062] The transmission group **50** has a support frame **51** (a.k.a., bracket, base, etc). The support frame **51** is used to interface the linear actuator system **10** to a motion platform. The motion platform may be a seat, a flat platform, a flat base, a plate, or any other suitable end effector, an example of which is given below. The support frame **51** may have a generally elongated shape having a plate **51**A. A pair of side walls **51**B project from the plate **51**A so as to define a receptacle with the plate **51**A, in which other components of the linear actuator system **10** will be received, including the actuator assembly as a whole as a possibility. Flanges **51**C may be provided at a top edge of the side walls **51**B so as to secure the support frame **51** to a motion platform. As observed, holes may be defined in the flanges **51**C so as to use standard fasteners as one possible way to secure the support frame **51** to a motion platform. For example, as shown in FIGS. **6**A-**6**C, the support frame **51** may be connected to an underside of the motion platform MP. In an embodiment, the whole actuator assembly is located entirely below the plane shown by MP, which plane may be coplanar with the underside of the motion platform MP. In another embodiment, the linear actuator system **10** is inverted with the flanges **51**C against the floor or structure, with the linear actuator system **10** moving the motion platform MP in the Y direction, such as generally upward.

[0063] In an embodiment, the support frame **51** is made from a monolithic metal plate that may be bent, cast, etc to have the receptacle shape described above. Other constructions (U brackets, saddles, box, etc) are possible as are other materials. The support frame **51** must have the structural integrity to support the actuator assembly and sustain the motions involved.

[0064] A pivot **51**D may be provided at an end of the support frame **51** and defines rotational axis A**1**. The pivot **51**A may be a single shaft as illustrated, or a pair of pins, a receptacle, or any other pivot component that will form a rotational joint with another member.

[0065] Still referring to FIGS. **1** to **3**, the casing of the motor group **20** is accommodated in the receptacle of the support frame **51**. However, it may be allowed to pivot as it transmits movement. Therefore, in an embodiment, the casing of the motor group **20** is supported by a pair of side plates **52** (a.k.a., links, swing members, etc) on opposite sides of the motor group **20**, as shown in FIGS. **4** and **5**. The side plates **52** may have pivots **52**A by which they are connected to the support frame **51**. The pivots **52**A may be a single shaft, a pair of shaft portions as illustrated, or a pair of pins, receptacles, or any other pivot component that will form a rotational joint with another member. The pivots **52**A are in line with one another and define rotational axis A**2**. Fasteners **52**B may be provided on the side plates **52** in order to attach the casing **21** of the motor group **20**—or any other part of the actuator assembly—to the side plates **52**. The fasteners may for example be bolts, set screws, etc. Other fastening configurations are considered, including welding, etc. In an embodiment, the side plates **52** may be integrally formed into the casing **21** as one possibility.

[0066] Referring to FIGS. **4** and **5**, a movement interface **53** is operatingly connected to the support frame **51** and may interface the transmission group **50** to a floor or to a motion platform. The movement interface **53** may have a pair of arms **53**A that are connected to the support frame **51** by way of the pivots **52**A, though independently of the movement of the side plates **52**. Therefore, both the motor group **20** and the movement interface **53** rotate about axis A**2**. However, it is also possible to have the axis of rotation of the movement interface **53** being offset from axis A**2**. For example, axis A**2** for the motor group **20** may be farther away that the axis of rotation of the movement interface **53** relative to axis A**1**, to increase the magnitude of movement of the movement interface **53**, for example. The free ends of the arms **53**A may be joined by a central member **53**B. In an embodiment, the arms **53**A are elongated and form a monolithic piece with the central member **53**B, although this is not necessary. The movement interface **53** may consequently have a U shape, or swing shape. The arms **53**A may pivot, but as they are essentially elongated components, the movement of the central member **53**B is quasi-translational, i.e., along direction Y, and thus generally vertical. A joint member **53**C may be provided in the central member **53**B in order to connect a floor pad, a base, a caster, etc. to the movement interface **53**. As shown in FIGS. **6**A to **6**C, the joint member **53**C may be in the form of a sphere (or complementary spherical receptacle) so as to form a spherical joint with a base that could be located on the floor or against the motion platform MP. This is one possible configuration among others. The free ends of the arms **53**A further define pivots **53**D that have a pivot axis A**3**.

[0067] A piston bracket **54**, or equivalent connector component, is located at an end of the piston of the driven group **40**. Therefore, the piston bracket **54** may translate upon actuation from the motor group **20**, in that the piston bracket **54** may be connected to a shaft or a piston of driven group **40**. Therefore, the piston bracket **54** moves in translation but may also rotate slightly due to the rotational mount of the motor group **20** to the support frame **51** via the pivot axis A**2** of the side plates **52**. A direction of the translation, along X, is essentially transverse to the various axes A**1**, A**2** and A**3** described above. The piston bracket **54** may be in the form of a U-shaped bracket (e.g., a clevis portion) having a pair of pivots **54**A, defining an axis of rotation A**4**. The pivot **51**A may be a single shaft, a pair of pins, a receptacle, or any other pivot component that will form a rotational joint with another member.

[0068] Cams **55** are responsible for converting the movement of the piston bracket **54** in direction X to a vertical movement in direction Y of the central member **53**B of the movement interface **53**. The expression "cam" is used as the joining link rotates and results in a generally translational movement in direction Y (though the movement may be more accurately described as being an arc of a circle). Although a pair of cams **55** is shown, a single cam could also be used. The cams **55** are pivotally mounted to the support frame **51** by the pivot **51**D. Therefore, the cams **55** rotate about axis A**1**. Moreover, the cams **55** are pivotally connected to piston bracket **54** at pivots **54**A, whereby the cams **55** rotate about axis A**4** relative to the piston bracket **54**. As a consequence of the cams **55** being pivotally connected to the support frame **51** at axis A**1**, and to the pivots **54**A of the piston bracket **54**, pivots **55**A at a free end of the cams **55** therefore move generally along the Y direction as a function of being pushed or pulled by the piston bracket **54**. The pivots **55**A define pivot axis A**5**. The pivots **55**A may be a single shaft, a pair of pins, a receptacle, or any other pivot component that will form a rotational joint with another member. Moreover, although the pivots **55**A are shown as offering only a rotation degree of freedom, it is contemplated to add a translational degree of freedom at the interface between the cams **55** and the links **56**. This may be achieved by having the pivots **55**A received in guide slots in the links **56**, as a possibility.

[0069] In the cams **55**, the axes A**1**, A**4** and A**5** are in a triangular parallel arrangement, to cause this Y-direction movement. The pivots **55**A could be locked elsewhere on the cams **55** to impart a different direction of movement to the pivots **55**A.

[0070] Links **56** interconnect the pivots **55**A of the cams **55** to the pivots **53**D of the movement interface **53** in direction Y. The links **56** may be required due to the fact that the pivots **55**A of the cams **55** have some remaining translation component in direction X with the push and pull action from the actuator assembly.

[0071] FIGS. **6**A, **6**B and **6**C show different moments of the displacement of the central member **53**B of the movement interface **53** as a result of the translational output from the actuator assembly of the linear actuator system **10**. With the actuator assembly being generally horizontal, it is seen that the output of the central member **53**B is essentially vertical. Moreover, the triangular arrangement of the axes A**1**, A**4** and A**5** may be equilateral in an embodiment, or near equilateral. This may cause the stroke of movement of the piston bracket **54** to have a value equal or close to a distance of movement of the central member **53**B. Other arrangements are possible, to amplify or reduce movements from the actuator assembly. In an embodiment, axes A**1**-A**5** are all parallel to one another to reduce the risk of mechanical jamming.

[0072] The embodiment of the transmission group **50** shown in the figures may have a twin set up, in that many of its components may be duplicated and/or may have a symmetry plane. In an embodiment the symmetry plane incorporates directions X and Y. The twin set up allows the forces on components to be spread, and may make the transmission group **50** more robust than without such a twin set up.

[0073] Referring to FIGS. **12**A to **12**E. various arrangements of the linear actuator system **10** are shown, using the axes A**1**-A**5** as reference, and with like reference numerals indicating a correspondence between the variant of FIGS. **1** to **6**, and the ones of FIGS. **12**A to **12**E.

[0074] In FIG. **12**A, the link **56** is connected to the movement interface **53** in such a way that axis A**3** is lower than the axis A**5**. The linear actuator system **10**, in the same manner as in FIGS. **1** to **6**, is not solidary to either one of the end points, i.e., the support frame **51** and the movement interface **53**. The arrangement of FIGS. **1** to **6** and of FIG. **12**A has six solidary parts: the structure of the linear actuator system **10**, the driven group **40**, the support frame **51**, the movement interface **53**, the cam **55** and the link **56**. The arrangement may feature six degrees of freedom of rotation, including the two on the axis A**2**, i.e., between the linear actuator system **10** and the support frame **51**, and between the support frame **51** and the movement interface **53**.

[0075] In FIG. **12**B, the structural portion of the linear actuator system **10** is fixed to the support frame **51**. Accordingly, piston bracket **54′** has two distinct pivot axes, A**4** and A**4′**. The arrangement of FIG. **12**B has six solidary parts: the structure of the linear actuator system **10** and support frame **51**, the driven group **40**, the movement interface **53**, the piston bracket **54′**, the cam **55** and the link **56**. The arrangement may feature six degrees of freedom of rotation, at axes A**1**, A**2**, A**3**, A**4**, A**4′** and A**5**.

[0076] In FIG. **12**C, the structural portion of the linear actuator system **10** is also fixed to the support frame **51**. Cam **55′** is not rigidly connected to the piston of the driven group **40** or to the movement interface **53**, relying instead on sliding abutments to transmit movement from the linear actuator to the movement interface **53**. The abutment ends of the piston of the driven group **40** and/or to the movement interface **53** may have rounded contact surfaces to ease the transmission of movement. Other configurations are contemplated, including using low friction materials. Gravity may hold the components assembled and in contact. The arrangement of FIG. **12**C has four solidary parts: the structure of the linear actuator system **10** and support frame **51**, the driven group **40**, the movement interface **53**, and the cam **55′**. The arrangement may feature two rotational joints, at axes A**1** and A**2**, with two friction planes that are uncaptured. Captured sliding arrangements are also considered, such as a pin and slot mechanism to maintain contact between cam **55′** and at least one of driven group **40** and interface **53′**, as a possibility among others (roller in groove, etc).

[0077] In FIG. **12**D, the structural portion of the linear actuator system **10** is also fixed to the support frame **51**. The piston bracket **54′** has two distinct pivot axes, A**4** and A**4′**. Cam **55** has a sliding component **55″** to transmit movement from the linear actuator system **10** toward the movement interface **53**. The linear actuator system **10′** may be slidingly disposed on the movement interface **53**, via abutment **51′**. Consequently, the movement interface **53** is not driven to move by the linear actuator system **10**. Though referred to as movement interface, item **53** may essentially be the ground, or a support surface. The sliding component **55″** may be a cylindrical component (e.g., semi-cylindrical), a spherical or hemi-spherical component, or a roller(s) (wheel(s), caster(s)) as in FIG. **12**E. Gravity may hold the components assembled and in contact. Accordingly, piston bracket **54′** has two distinct pivot axes, A**4** and A**4′**. The arrangement of FIG. **12**D has four solidary parts: the structure of the linear actuator system **10** and support frame **51**, the driven group **40**, the piston bracket **54′**, and the cam **55′**. The arrangement may feature three rotational joints, at axes A**1**, A**4** and A**4′**, with one uncaptured friction plane, with high friction for the assembly to remain in position.

[0078] The arrangement of FIG. **12**D has four solidary parts: the structure of the linear actuator system **10** and support frame **51**, the driven group **40**, the piston bracket **54′**, and the cam **55**. The arrangement may feature three rotational joints, at axes A**1**, A**4** and A**4′**, with one uncaptured friction plane, with high friction for the assembly to remain in position.

[0079] The arrangement of FIG. **12**E has four solidary parts: the structure of the linear actuator system **10** and support frame **51**, the driven group **40**, the piston bracket **54′**, the cam **55**, and the roller **55′**. The arrangement may feature three rotational joints, at axes A**1**, A**4**, A**4′**, and A**7** at the wheel **55′**.

[0080] Based on FIGS. **1** to **6** and **12**A-**12**E, the linear actuator system **10** may be generally described as having an actuator assembly for moving an output in translation in a first direction A transmission has a frame. A joining link(s) is pivotally connected to the frame at a first location (e.g., axis A**1**) and operatively connected to the actuator assembly at a second location (e.g., axis A**4**, axis A**4′**) for receiving movement from the output. The joining link(s) contacts an interface at a third location (e.g., axis A**3**, axis A**5**) to cause relative movement between the frame (e.g., **51**) and the interface (e.g., **53**) in a second direction, such as Y, differing from the first direction, such as X.

[0081] With reference to FIGS. **7-11**, a motion platform system according to another aspect of the present technology and generally shown at **60** will now be described. Although one may use the motion platform system **60** by simply standing thereon, the motion platform system **60** is well suited to be paired with seating of a conventional type such as a chair, an office chair, or a more specialized type such as a gaming chair or a pilot seat found in a simulator. One such seat may, as it will be appreciated from the forthcoming description, be rendered an end effector for transmitting an output of the motion platform system **60** to a user of the seat. Indeed, the motion platform system **60** may impart the seat with movements in synchrony with one or more signals that may include video, sound and/or a signal indicative of an input device being used, for example a controller connected to a simulator or gaming system. Such movements may be devised to impart the seat with vibro-kinetic effects of an amplitude suitable for simulating movement and haptic events, as the case may be. The motion platform system **60** therefore produces a motion output that imparts a variation in position and/or orientation of the seat that may be defined, at least in part, relative to one or more of directions Px, Py, Pz of a reference coordinate system P. In the depicted embodiments, the motion output includes a component along the direction Pz, in this case not purely translational (though it could be), but rather coupled to rotational movement about a rotation axis R parallel to the direction Py. In embodiments, the motion output may include any combination of translational component(s) and rotational component(s) to impart the seat with desired motion via the motion platform system **60**. The movements may be for example described as pitch and roll. The number of degrees of freedom (DOF) of movement may vary depending on the nature of the motion platform system **60**. For example, the motion platform system **60** of FIG. **6** with a single one of the linear actuator system **10** may output a single DOF of movement.

[0082] The motion platform system **60** includes a support structure **70**, a number of joints **80** (FIG. **8**A), **90** (FIG. **8**B) and an actuator connected to the support structure **70**, as well as a motion structure **100** kinematically coupled to the support structure **70** via the joints **80**, **90** and the actuator so as to govern the motion output of the motion structure **100** of the motion platform system **60**. The actuator is arranged to output a vertical movement, i.e., such that its output occurs at least to some degree along the direction Pz. As in the embodiment depicted in FIGS. **7-10**, the actuator may correspond to the actuator system **10** described hereinabove, here shown laid over the support structure **70** such that its directions X and Y generally correspond to the directions Px and PZ, respectively. The actuator system **10** may be arranged such that its movement interface **53** connects to the support structure **70** via a first joint **80** from the joints **80**, **90** (referred henceforth as the first joint **80**). The actuator system **10** could also be laid on the ground or on the support structure **70**, for instance for arrangements such as in FIGS. **12**D and **12**E. Because of the arrangement described below, the motion platform assembly system **60** may be of low profile, such as at most 12 inches in height from the support structure to a support plane, or top plane, of the motion structure **100**. The first joint **80** may be a ball joint. The first joint **80** may be constrained relative to the support structure **70**, whether fixedly, translatably or rotatably so. Although the first joint **80** is described as a piece provided in addition to the actuator system **10**, the first joint **80** may in some implementations form an end piece of the movement interface **53**. Other types of joints may be between the movement interface **53** and the support structure **70**, such as a universal joint. In some implementations, the first joint **80** may include more than one component, for example components each provided with one or more DOFs, such as a slider connected to the support structure **70** and a pivot connected between the slider and the actuator system **10**, or components provided with a sole, common DOF, such as pivots disposed in a hinge-like arrangement. A rigid connection is also contemplated, for instance with additional DOFs being provided at other joints to enable the movement between the support structure **70** and the motion structure **100**.

[0083] Second joints **90** from the joints **80**, **90**, link the motion structure **100** to the support structure **70** (e.g., a panel) at a location spaced away from the first joint **80** in a plane incorporating Px and Py. Although it could consist of a single panel, the motion structure **100** has a pair of opposite sides **110**, **120**, or portions, or plates, or panels (first panel, second panel), of which a first, support-facing side, or actuation portion **110**, interfaces the actuator system **10**, for example via the support frame **51**. The sides **110** and **120** are shown as panels of metal sheeting, but other panels could be used, such as a molded honeycomb structure. A second side or output portion **120** of the motion structure **100**, also referred to as a support platform, bears docking features **130**. As will be described hereinafter, the docking features **130** include at least a concavity to receive a wheel in an embodiment. The docking features **130** may include one structural feature of the motion structure **100** defining a non-permanent yet secure attachment means that is suitable for operatingly connecting the motion platform system **60** to any one of a wide range of seats, or for supporting a user standing on it. A sleeve **140** (FIG. **11**), bellows-like member or other membrane-like component may be attached to peripheral edges of the support and motion structures **70**, **100** to follow relative movements thereof, shielding internal components of the motion platform system **60**. Components such as footrests, speakers, input devices and other implements may be provided on the motion structure **100**.

[0084] In view of FIG. **8**, it will be appreciated that the support structure **70** is a ground-interfacing component of the motion platform system **60** provided as a spatial reference relative to which other components of the motion platform system **60** may be positioned. The support structure **70** may optionally be suitably arranged for levelling, protecting and/or spatially arranging other components relative to one another. Alternatively, the motion platform system **60** may be without the support structure **70**, and lay instead on the ground or on a surface of a structure. In the depicted implementation, the support structure **70** is constructed of sheet metal having been cut and shaped into a plate-like structure having a flat bottom **72** surrounded with a peripheral wall **74**. The support structure **70** may support, or otherwise hold in position, either the joints **80**, **90** and/or the actuator system **10**. Tabs **74**A, holes **76**A, groove **76**B in one of the tabs **74**A or other like features may be present on the support structure **70** to this end. The sheet metal structure shown is one possibility among others. As an alternative or as an addition, it is contemplated to use a framing structure as another possibility, with the framing structure being made of elongated beams interrelating the various components as set out above.

[0085] The actuator system **10**, the first joint **80**, and the second joints **90** may be described as a motion-governing group of the motion platform system **60**. Each one of the joints **80**, **90** is independently fixedly attached relative to the support structure **70**, although this is merely one possible implementation among those contemplated. A single joint **90** could be used if a single actuator system **10** is present.

[0086] Turning to FIG. **8**A, the particular, exemplary implementation of the first joint **80** of the present embodiment will be described in greater detail. The first joint **80** is connected to an end of the actuator system **10** (e.g., the movement interface **53**) whereas an opposite, relatively displaceable end of the actuator system **10** (the support frame **51**) is fixed to the motion structure **100**. Hence, the first joint **80** imposes that motion of the movement interface **53** relative to the support structure **70** be limited to rotation about three axes. The first joint **80** may be a ball joint having a first support member **82** connected to the support structure **70**, and a first output member **84** connected to the actuator system **10**. The first joint **80** may be constrained relative to the support structure **70**, whether fixedly, translatably or rotatably so, for example by way of a bracket **86**. In some such implementations, one of the first support member **82** and the first output member **84** may be integral to the movement interface **53** which, for instance, may define a socket of the ball joint, or a ball of the ball joint. The first support member **82** is a base **82**A defining a socket **82**B, and the first output member **84** includes a stem **84**A and a ball **84**B joined thereto. It should be noted however that implementations of the first joint **80** in which the first support member **82** and the first output member **84** respectively define the ball and the socket of the ball joint are contemplated. On the outside, a bottom of the base **82**A interfaces the support structure **70**. On the inside, the base **82**A defines a cavity surrounding components that define the socket **82**B. Such socket-defining components may in some implementations be slidable relative to the base, at least to some degree, with the output side member **84** whose ball **84**B is received therein. The first joint **80** may be provided with a boot attached to peripheral edges of the base **82**A and of the stem **84**A to follow relative movements thereof, shielding internal features of the first joint **80**. In this embodiment, the bracket **86** is provided as one of various suitable means for holding the first joint **80** relative to the support structure **70**, namely to restrain motion of the first support member **82**. The bracket **86** may be fastenable to the support structure **70**, for example via the groove **76**B. A plate-like top portion **86**A of the bracket **86** may be said to cover at least a portion of the first support member **82**, via which the bracket **86** may limit or block movement of the first support member **82** relative to the support structure **70** in the Pz direction. In this implementation, the top portion **86**A of the bracket **86** includes projections **86**B extending on either side of the first support member **82** to hinder its movement in the Py direction. The top portion **86**A may define an opening **86**C through which the first output member **84** may extend. The opening may be sized so as to allow a suitable range of motion to the first output member **84** as it moves relative to the first support member **82** or even as it slides therewith relative to the bracket **86**. In some implementations, some minimum translational movement of the first joint **80** may be possible to lessen stresses on the first joint **80**.

[0087] Referring to FIG. **8**B, it may be observed that the second joints **90** are pivots spaced from one another in a direction corresponding to that of the Py direction, though this is merely an option. The second joints **90** (one of which is exploded) each have a second support member **92** fixed to the support structure **70**, and a second output member **94** fixed to the motion structure **100**, namely its actuation portion **110**. The second joints **90** may be similar in shape and in function, and may be disposed in line with one another in a hinge-like arrangement so as to define a common rotation axis R. The rotation axis R may be parallel to the Py direction. As the second joints **90** concurrently define a common rotation axis R, they constrain movement of the motion structure **100** relative to the support structure **70** in one rotational DOF. Therefore, although they are described as a pair of second joints **90**, as they are discrete items, the second joints **90** may also be referred to as a single joint constraining movement to a single rotational DOF.

[0088] In the second joints **90**, each second support member **92** may include a housing **92**A with an inner diameter mounted to a pin-like shaft **92**B, for example by way of a bearing. The bearing may be a spherical bearing, among other possibilities. Each second output member **94** may have a pair of prongs **94**A merging together into a socle **94**B, so as to define an inverted U shape therewith. A throughbore through the prongs **94**A, may be sized to receive the shaft **92**B. In this implementation, the housing **92**A is pierced between the prongs **94**A such that the inner diameter of the housing **92**A aligns with an inner diameter of the bore, so that the shaft **92**B may extend therethrough, supported by the housing and supporting the second output member **94** on either side of the housing **94**A. In this implementation, the second output member **94** may be said to be mounted directly to the shaft **92**B, as an outer diameter of the shaft **92**B corresponds to the inner diameter of the bore. In other implementations, the second output member **94** may be mounted indirectly to the shaft **92**B via one or more bearings fitted to the bore. The second output member **94** is mechanically joined to the motion structure **100**, in this case via a connector **94**C fastened to the socle **94**B from across the actuation portion **110**. The connector **94**C may be a disk with a fastener, also described as a flanged bushing have a first, narrow end extending through an opening of the actuation portion **110** to be lodged into the socle **94**B, and a second, wider end resting against the actuation portion **110**. This means of joining the second joints **90** to the motion platform **100** may desirably distribute mechanical stress and mitigate loosening or wear of interfacing components, for instance by way of the disk increasing a contact surface between the second joint **90** and the motion platform **100**. The second joint **90** as provided in certain other embodiments may differ functionally (e.g., provide additional degrees of freedom) and/or structurally. For instance, the second joints **90** may be provided in the form of a sole second joint such as a pivot, extending axially (i.e., in an orientation parallel to the rotation axis R) between opposite ends respectively disposed on opposite sides of the actuator system **10**. The one or more second joints **90** may also be structured to allow other degrees of freedom in addition to rotation about the R axis, for example rotational movement about an axis that is orthogonal to the R axis, or even translational movement. For example, a pair of ball joints may be used. The second joints **90** may extend to either side of a notional plane in which the first joint **80** and the X direction of the actuator system **10** lay. Respective projections of the rotation axis R and of the X direction of the actuator system **10** in the plane of the Px and Py directions may be orthogonal.

[0089] The support members of the first and second joints **80**, **90** are indirectly bound to one another so as to be held in a common position relative to a plane in which lay the Px and Py directions as the motion platform system **60** operates. Stated otherwise, the support members of the joints **80**, **90** are fixedly connected to the support structure **70** at respective positions so as to maintain a common spatial relationship. The foregoing represents one non-limiting, exemplary spatial arrangement of the joints **80**, **90** which may desirably distribute and balance loads imparted via the motion structure **100** to the joints **80**, **90** and to the actuator system **10** as the motion platform system **60** operates.

[0090] The actuator system **10** is typically operated via a controller which may, for example as in the embodiment of FIG. **8**, be provided as a part of the motion platform system **60**. Generally shown at **10**A, the controller **10**A may be integrated with a power supply and packaged into a housing disposed proximate the actuator system **10**. The controller **10**A may even be sized and arranged so as to be shielded by other components of the motion platform system **60**, for example by the bottom **72** of the support structure **70**.

[0091] In the embodiment depicted in FIGS. **7-10**, the actuator system **10** is mounted so as to extend lengthwise between the joints **80**, **90**, and oriented such that its movement interface **53** generally faces toward the support structure **70**, namely its bottom **72**. Conversely, in this orientation, the support frame **51** of the actuator system **10** generally faces away from the support structure **70** and toward the actuator-facing portion **110** of the motion structure **100**. The movement interface **53** is operatingly connected to the support structure **70** via the first joint **80**, and the support frame **51** is secured to the motion structure **100** via the actuator-facing portion **110**. Consequently, actuation of the actuator system **10** causes a relative movement between the support structure **70** and the motion structure **100**.

[0092] The support structure **70** and the motion structure **100** may be similarly sized such that, in use, the motion structure **100** generally remains above the support structure **70**. A portion of the motion structure **100** may even overlay the controller **10**A opposite the support structure **70**. Moreover, a footprint of the motion platform system **60** may be shaped so as to correspond to that of a chair to be used therewith. In the present embodiment, the footprint (i.e., the contour of the motion structure **100**, but also of the support structure **70**) is pentagonal in shape and sized to match a footprint of a five-prong chair base. Other shapes are possible, such as circular, square, etc, whether or not as a function of the number of legs. In other implementations, either one or both of the support structure **70** and the motion structure **100** may be a web-like arrangement of interconnected members suitably arranged for connecting to the joints **80**, **90** and to the actuator system **10** in a manner consistent with the foregoing. The motion structure **100** may in certain cases overhang from the support structure **70** and above the ground (and the support structure **70** could be the ground as well). The motion structure **100** may be constructed of sheet metal having been cut and shaped into a plate-like structure. In embodiments, the actuation portion **110** and the output portion **120** are distinct, plate-like structures together forming the motion structure **100**. The actuation portion **110** (or first plate **110**), has a generally flat bottom **112** surrounded with a peripheral edge wall **114**. Tabs **114**A may project from the peripheral edge wall **114**. The sleeve **140** may be affixed to the motion structure **100** via such tabs **114**A and, similarly, to the support structure **70** for example via the tabs **74**A. Openings **116**A and cutouts **116**B may be formed in the first plate **110**. For example, the second output members of the second joint **90** and the support frame **51** of the actuator system **10** may be fastenable to the motion structure **100** via such openings **116**A. The cutouts **116**B, on the other hand, may provide clearance between the motion structure **100** and other nearby components. The actuator system **10** may be fitted to one such cutout **116**B such that a distance between the motion structure **100** and the support structure **70** may be less than a height of the actuator system **10** measured along its direction Y. The output portion **120** (or second plate **120**) of the motion structure **100** also has a generally flat bottom **122** surrounded with tabs **124**A projecting from a peripheral edge **124**. In the depicted embodiment, the tabs **114**A and **124**A are complementarily staggered, with the tabs **124**A overlapping the walls **114**. Landforms **126** (e.g., cutouts such as slits **126**A, troughs **126**B, holes **126**C, or even embossing) of various shapes and sizes may be formed in the second plate **120**, some or all of which may be part of the docking features **130** of the motion structure **100**. Casters of a chair may be received directly in the troughs, with an axis of rotation of the casters being parallel to lateral edges **128**B projecting from an edge **128**A of the cutouts **126**B. Such straight cutouts may receive wheels or casters of different diameters, in such a way.

[0093] Referring to FIGS. **9** and **10**, characteristics pertaining to relative positioning of components of the motion platform system **60** will be detailed. In FIG. **9**, a lowermost position of a motion range of the motion platform system **60** is shown, corresponding to a lowermost position attainable by any portion of the motion structure **110**. The lowermost position may also be described as the position in which a distance taken along the Pz direction from the motion structure **100** to the bottom **72** of the support structure **70** is minimized. In the present embodiment, the lowermost position is attained upon the bottom **122** of the second plate **120** abutting against a portion of the movement interface **53** opposite that connected to the first joint **80**. In contemplated variations, the motion platform **100** may be shaped such that its motion is absent hindrance by any singular contact that may otherwise occur throughout the motion range. In the lowermost position, the motion structure **100** is not parallel to the support structure **70**, as the distance between the support structure **70** and the motion structure **100** at the second joints **90** is greater than at the first joint **80**. Hence, in this position, the first output member **84** of the first joint **80** may be pivoted relative to its corresponding support member **82**, with the movement interface **53**, by a first initial angle αo about the axis A**1**. The first initial angle αo may for example be counter-clockwise when observed in the plane of FIG. **9**. Still in the lowermost position, the second output members **94** of the second joints **90** may be pivoted relative to their corresponding support members **92**, with the motion structure **100**, by a second initial angle βo about the rotation axis R. The second initial angle βo may for example be clockwise when observed in the plane of FIG. **9**. Likewise, a seating axis S corresponding to a vertical orientation of a seat connected to the motion platform system **60**, may thus be pivoted with the motion structure **100** at the angle βo. As will become apparent from the forthcoming, the location of the seating axis S with respect to the motion structure **100** is generally determined by the docking features **130**. Also, the location of the joints **80**, **90**, their respective initial angles αo, βo and various dimensional and structural considerations of the motion platform system **60** may be determined as a function of spatial and loading characteristics of the seat, which may differ based on the type of the seat being used, on user build and weight, and even on user preferences. In the depicted embodiment, it shall be noted that the first and second joints **80**, **90** are disposed on either side of the seating axis S with respect to the direction Px, with the rotation axis R being closer than the axis A**1** of the actuator system **10** with respect to the seating axis S—however other arrangements are possible. The above configuration may assist in effectively transmitting loads between the seat and the actuator system **10**, for instance by rendering the output of the actuator system **10** as felt by a user via the seat appear vertical (or aligned with the seating axis S), and/or by compensating for an offset of a center of gravity of the user and the seat, typically forward (i.e., transversely away from the seating axis in the Px direction) due to users leaning toward a front of the seat toward display devices, input devices or the like.

[0094] As indicated hereinabove, the docking features **130** provide one or more attachment points for seating to be securely connected to the motion platform system **60**. In FIG. **10**, an exemplary implementation of the docking features **130** particularly suited for securing a telescopic chair will now be described in greater detail. The docking features **130** may include a plurality of groups of docking features **130**A that may be distributed on the output portion **120** of the motion structure **100**, for example along its periphery and/or near or at a center of the second plate **120**. The groups of docking features **130** may be disposed in circumferential and/or radial positions relative to the seating axis S, in some cases evenly so, forming a pattern. In this embodiment, the docking features **130** includes seating fasteners **132** angularly spaced from one another relative to the seating axis S. A total of five seating fasteners **132** are disposed 72 degrees from one another, and respectively spaced radially inwardly from the periphery of the motion structure **100** relative to the seating axis S, a five-prong configuration reflecting that commonly found in chairs supported by five-prong wheeled bases. Some chairs may have three prongs, four prongs, etc, and the motion structure **100** may be adapted for such chairs as well. The seating fasteners **132** are in this case straps **132**A attached to the motion structure **100** via the slit-like cutouts **126**A defined in the second plate **120**, adjustable in size via a buckle **132**B (alternatives being Velcro®, snap connectors, elastics (e.g., bungees), etc). This implementation of the seating fasteners **132** is adapted to conform to various shapes of chair base prongs, which may differ from one another whether in length and/or in cross-sectional shape. In alternate implementations, seating fasteners **132** may allow to fasten a chair via its wheels, or even via sockets of the prongs via which the wheels pivotally connect. The docking features **130** of the present embodiment includes wheel fitting features, in this case disposed radially inwardly of the periphery of the motion structure **100** and radially outwardly of the seating fasteners **132**. In this particular arrangement, each one of the wheel fitting features is in radial alignment with a corresponding one of the seating fasteners **132**. The wheel fitting features may be concave troughs, and may be said to be conformable to various size wheels and various radial arrangements thereof. Indeed, the wheel-fitting features have radially elongated contours, which may correspond, at least in part, to that of the suitably dimensioned through-like cutouts **126**B formed in the second plate **120**. In the present implementation, each one of the wheel-fitting features extends radially outwardly relative to the seating axis S from the innermost edge **128**A being an edge of a corresponding cutout **126**B to an outermost edge corresponding to the wall **114** of the actuation portion **110** of the motion structure **100**, the through-like cutouts **126**B being outwardly open-ended, e.g. flaring outwardly. The lateral edges **128**B of each wheel-fitting feature, also being edges of the corresponding cutout **126**B portions, extend lengthwise alongside one another between corresponding innermost and outermost edges **128**A, a space therebetween defining a width of the wheel-defining feature. Depending on their size and shape, wheels may be positioned differently to be received by a suitably aligned wheel-defining feature. For example, a wheel narrower than the width of the wheel-defining feature may be received such that it extends into the cutout **126**B with its sides extending generally radially relative to the seating axis S and respectively facing a corresponding one of the lateral edges **128**B. It shall be noted that the lateral edges **128**B offer tread-contacting, circumferentially-spaced supports for a virtually unlimited range of wheel diameters upon such wheels being suitably positioned with their sides in a radial orientation relative to the seating axis S. The shape of the lateral edges **128**B and the width of the wheel-fitting features may nevertheless be adapted to provide optimal support for a pre-determined range of wheel diameters. The wheel-fitting features may also be layered with materials, for example along one or more of their edges **128**A, **128**B, which may desirably assist wheel retention and/or mitigate wheel wear under normal use conditions. As another possibility, wheel-fitting features of various sizes may be offered, and selected as a function of wheel sizes. The docking features **130** also includes a stem-fitting feature, here provided in the form of the cutout **126**C, in this case circular in shape, defined in the second plate **120** and surrounding the seating axis S. This stem-fitting feature may be sized for receiving a bottom end of a telescopic stem of a chair which, under certain circumstances, may otherwise collide with the motion structure **100**. The docking features **130** may in some embodiments include features provided in different numbers, with different individual shapes, disposed in different patterns, and/or be added, removed or interchanged to customize support, clearance and even output transmission characteristics of the motion structure **100**.

[0095] In some embodiments, the actuator, joints and docking features of the motion platform system **60** may differ from those described hereinabove, whether in terms of kinematics and/or of form factor.

[0096] Turning now to FIG. **11**, one such embodiment of the motion platform system **60** provided with a vertically-oriented linear actuator system **10′** is shown. The motion platform system **60** may have numerous similar components as the one of FIGS. **7-10** whereby like components will bear like reference numerals. The actuator system **10′** includes a casing held by a support frame **51′**, a motor received in the casing, and a movement interface **53′** connected to the joint **90** (although a joint **80** could be used). The motor is operatingly connected to the movement interface **53′** such that it may be reciprocated along a translational direction Y′ relative to the casing. The support frame **51′** is mounted to the motion structure **100**, in this case on top of the second plate **120**, holding the casing such that the movement interface **53′** faces toward the support structure **70**. By this arrangement, the direction Y′ is at an angle relative to the plane in which the directions Px, Py lay. The movement interface **53′** may thus engage the support structure **70** via the joint **90** such that the motion structure **100** reciprocates pivotally about the axis R as the movement interface **53′** reciprocates along the direction Y′. A rotational axis R′ of the joint **90** at the movement interface **53′** may be transverse to the rotational axis R of the other joints **90** in FIG. **11**.

[0097] In embodiments, alternate implementations of the motion structure **100** may be provided. Still referring to FIG. **11**, the docking features **130** include wheel-fitting features disposed in an even, circular pattern relative to the seating axis S. Outer cutouts **126**B (characterized as outer relatively to the seating axis S and in contrast to inner cutouts **126**C, when present) have a closed contour defined in the second plate **122** at a location spaced inwardly from the peripheral edge **124**. The contour of each outer cutout **126**B defines innermost and outermost edges **128**A of the wheel-fitting features, and lateral edges **128**B extending therebetween, in this case the latter also defining pockets. Such pockets may, depending on the implementation, form retentive features or clearance for other components being part of the docking features **130**. Such components may be inserts **134** fitted to the outer cutouts **126**B and attached to the motion structure **100** via one of various suitable means. For example, the inserts **134** may have a shape complementary to that of the outer cutouts **126**B and structured so as to be mechanically retained by the second plate **120** upon being received by one of the outer cutouts **126**B. In the present embodiment, each insert **134** defines flanges **134**A overlaid onto the second plate **120**, namely on either side of the corresponding outer cutout **126**B, and held in place via fasteners. It should be noted that various means are contemplated for securing the inserts **134** to the motion structure **100**, some of which may be permanent such as chemical adhesives and welding.

[0098] Each insert **134** may define a recessed surface **134**B laterally flanked by the flanges **134**A and shaped so as to extend inwardly into the motion platform **100** via one of the outer cutouts **126**B upon its adjoining flanges **134**A laying against the second plate **120**. The recessed surface **134**B may extend radially outwardly relative to the seating axis S as it extends away from the inner edge **128**A. A cross-sectional profile of the recessed surface **134**B may be V-shaped as shown, or shaped otherwise to conform to a wide variety of wheel shapes. Alternatively, recessed docking features can be created by slots (or recessed V-shape surfaces) directly at the surface of the motion platform output plane **120**. For instance, five slot cutouts arranged at 72 degrees around center axis S could receive the five wheels of an office chair, the wheels being oriented perpendicular to the slots so that each wheel is immobilized by gravity and 2 contact points with the slot along the periphery of the wheel. Lateral edges **134**C may be defined either by the flanges **134**A, the recessed surface **134**B or, as in the depicted implementation, may correspond to a bend in the insert **134** formed where the recessed surface **134**B meets each of its adjoining flanges **134**A. Each insert **134** may include one or more seating fasteners **132**, here passing through slit-like openings **134**D of the insert **134**, defined in pairs opposite one another in the flanges **134**A. The seating fasteners **132** may be used to strap a prong or wheel of a chair disposed thereon to the underlying insert **134**, against the recessed surface **134**B, to secure the chair to the motion structure **100**. More than one pair of openings **134**D may be provided lengthwise between the innermost and outermost edges **128**A of any given wheel fitting feature, allowing to select which openings **134**D to use for fastening a certain type of chair and/or to use more than one seating fastener **132** for a given prong or wheel of a chair. The openings **134**D may line up with the pockets defined by underlying lateral edges **128**B or, in other implementations, with other suitably sized and positioned slit-like openings defined in the second plate **120**. In yet other implementations, the seating fasteners **132** are surface-mounted, meaning that they are secured to a remainder of the motion structure **100** without extending underneath any of the inserts **134** or the second plate **120**.

[0099] In embodiments of the motion platform system **60**, a plurality of actuators may be provided, suitably sized and joined relative to the motion structure **100** so as to impart desired degrees of freedom and ranges of motion thereto. In some such embodiments, a secondary actuator may be arranged to effect a secondary output targeting a portion of a chair secured to the motion structure **100**, for example a portion of the seat, a portion of the base or a wheel, to impart motion thereto in a distinct, radially offset manner relative to the output of the actuator as described hereinabove.

Claims

1. A motion platform system comprising: a support structure; a motion structure operatively mounted to the support structure by at least one joint so as to be displaceable relative to the support structure in at least one degree of freedom; and at least one of the linear actuator system including an actuator assembly for moving an output in translation in a first direction, and a transmission having a frame, at least one joining link pivotally connected to the frame at a first location and operatively connected to the actuator assembly at a second location for receiving movement from the output, the at least one joining link contacting an interface at a third location to cause relative movement between the frame and the interface in a second direction differing from the first direction, wherein the linear actuator system is operatively connected to the support structure and the motion structure, the linear actuator system actuatable to impart movement to the motion structure in the at least one degree of freedom.

2. The motion platform system according to claim 1, wherein the motion structure includes a first panel configured to define a motion platform.

3. The motion platform system according to claim 2, wherein the first panel has receptacles configured for receiving casters of a chair.

4. The motion platform system according to claim 3, wherein the receptacles are elongated troughs.
5. The motion platform system according to claim 4, including five of the elongated troughs, the elongated troughs being circumferentially distributed 72 degrees apart.
6. The motion platform system according to claim 3, including a strap for each receptacle.
7. The motion platform system according to claim 2, wherein the first panel has a pentagonal shape.
8. The motion platform system according to claim 2, including a second panel being located under the first panel, the linear actuator system being fixed to the second panel.
9. The motion platform system according to claim 8, wherein the at least one joint is connected to the second panel.
10. The motion platform system according to claim 8, wherein the second panel has a pentagonal shape.
11. The motion platform system according to claim 8, wherein the support structure is a third panel being located under the second panel.
12. The motion platform system according to claim 11, wherein the third panel has a pentagonal shape.
13. The motion platform system according to claim 8, wherein the support structure is the ground.
14. The motion platform system according to claim 1, including a spherical joint between the linear actuator system and the support structure.
15. The motion platform system according to claim 1, wherein the at least one joint has a pivot.
16. The motion platform system according to claim 1, wherein the at least one joint includes two pivot members spaced apart and sharing a common rotational axis.
17. The motion platform system according to claim 1, wherein a height between the support structure and a support plane of the motion structure is at most 12 inches high.
18. A linear actuator system comprising: an actuator assembly having an output, the actuator assembly actuatable for moving the output in translation in a first direction; and a transmission having a frame, at least one joining link pivotally connected to the frame at a first location and operatively connected to the actuator assembly at a second location for receiving movement from the output, the at least one joining link configured to contact an interface at a third location; wherein the transmission causes relative movement between the frame and the interface in a second direction differing from the first direction when receiving movement from the output.